

1

Multiple choice 1 point Question

What makes topic detection more difficult for longer text compared to short text?

- ☒ Longer texts such as news may contain different topics which creates ambiguity.
- ☐ Longer texts such as news have stronger opinions than shorter texts
- ☐ Longer texts have more variation in words to express topics
- ☐ Longer texts provide more details than shorter texts about a topic

2

Multiple choice 1 point Question

Why is it so difficult to agree on a set of universal topics?

- ☒ Topics are subjective categories and both the world and our view on the world changes continuously.
- ☐ There is a limited set of topics but there are many different ways to describe them.
- ☐ Topics are based on the keywords that are different from text to text.
- ☐ There is too much variation in the keywords that can be extracted from topics.

3

Multiple choice 1 point Question

What is the most important difference between topic modelling and text categorisation?

- ☒ For text categorisation the topics are given whereas topic modelling does not assume an a priori definition of topics.
- ☐ Topic modelling is hierarchical and text categorisation is not.
- ☐ Text categorisation uses key words and topic modelling uses complete texts.
- ☐ Topic modelling is more fine-grained than text categorisation.

4

Multiple choice 1 point Question

What determines the level of granularity when assigning topics to a text?

- ☐ Topics should be fine-grained to capture all details
- ☐ Topics should be coarse-grained to capture only main topics
- ☒ Depends on data usage and specific application
- ☐ The number of topics is always fixed and does not vary

5

Multiple choice 1 point Question

At what unit level can a topic be assigned in topic modeling or classification?

- ☐ Complete books
- ☐ Short texts
- ☐ Sentences
- ☒ All of the above

6

Multiple choice 1 point Question

Which of the following is **not** a disadvantage of supervised topic classification?

- ☐ New topics that are not in the training data cannot be detected
- ☐ The results can be biased towards the distribution of training data
- ☐ Manual effort is required to maintain the topics and training data
- ☒ Control over the topics

7

Multiple choice 1 point Question

What approach is commonly used to label clusters in topic modeling?

- ☐ Random words from each cluster are chosen as labels
- ☐ Domain experts manually assign labels based on the most relevant terms in each cluster
- ☒ Extracting the most distinctive keywords from the clusters using techniques like TF-IDF
- ☐ Using the most frequent words across all clusters as labels for each individual cluster

8

Multiple choice 1 point Question

Why is random initialization and internal sampling considered a limitation in Latent Dirichlet Allocation (LDA)?

- ☐ It makes LDA computationally inefficient
- ☒ It can lead to unstable topic modeling results
- ☐ It leads to overfitting, making the model less generalizable
- ☐ Random initialization and internal sampling are not considered a limitation in LDA

9

Multiple choice 1 point Question

What is catastrophic forgetting during fine-tuning?

- ☒ The tendency of a model to forget information learned during pre-training when exposed to new data during fine-tuning
- ☐ The process of enhancing model performance by learning new tasks during fine-tuning
- ☐ The ability of a model to learn multiple tasks during fine-tuning
- ☐ The ability of a model to remain effective with little data available for fine-tuning

Which of the following factors is **not** mentioned as one to be considered when choosing a topic model, according to Churchill & Singh (2021)?

- ☐ Document length
- ☐ Noise level
- ☐ Dataset size
- ☒ The complexity of the algorithm